University of Moratuwa

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TECHNICAL REPORT

Sri Lankan Vehicle Listing Price Prediction

Model development, evaluation and deployment

IT5514 - Applied Machine Learning

Submitted By

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Abstract

This study evaluates seven regression approaches for predicting historical advertised vehicle prices in Sri Lanka, including NGBoost probabilistic regression alongside Random Forest, XGBoost, CatBoost, LightGBM, Ridge and an RF–XGBoost stack with a Ridge meta-model. The dataset contains 9,788 listings collected between December 2024 and February 2025. Training-only exploration, temporal validation and feature-group purging reduce leakage. Random Forest achieves the lowest validation MAE of 7.15 lakhs and test MAE of 5.83 lakhs. NGBoost learns a Normal price distribution and achieves test MAE of 8.58 lakhs. Its nominal 90% intervals cover 89.9% of historical test prices, but their average width is 33.61 lakhs and 2.4% have negative lower bounds. A Streamlit application supports model switching and inspection of predictions, diagnostics and explanations. The previously examined test set makes the seven-model comparison exploratory; confirmation requires newly collected later listings.

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Appendix A provides model-specific results and diagnostic figures.

1. Objective and local relevance

Predict historical advertised vehicle prices from tabular listing characteristics. Buyers and sellers can explore how vehicle characteristics relate to asking prices, while users can compare models and inspect their limitations. It does not estimate verified transaction prices or current 2026 market values. The study also examines whether NGBoost can provide useful distributional information alongside the point estimates, using interval coverage, width and distributional limitations as supporting evidence.

2. Dataset and provenance

Prasad Nirmal, SriLankan Vehicle(Car) Price Dataset, version 3. Publisher metadata identifies Ikman.lk and Riyasewana.com as sources and lists MIT licensing. Original CSV and metadata are retained in data/raw.

SHA-256: 4208d2673cf4d9c8d3d6042dc8a1b97004671812441975f5879550059fbce1ab. There are 9,788 source rows, spanning 2024-12-03 to 2025-02-05. Price is in LKR lakhs, confirmed by the project owner; 1 lakh is LKR 100,000.

3. Exploratory data analysis

Exploration in scripts/analyze_data.py uses only the training partition. The training price median is 42.5 lakhs, 95th percentile 120.0, maximum 790.0, and skewness 3.80. The long upper tail motivates reporting RMSE and price-band errors alongside MAE. Extreme observations remain in evaluation rather than being removed to improve scores.

Feature	Distinct values	Categories with fewer than 3 rows	Rows in rare categories
brand	48	14	18
model	1061	758	893
gear	2	0	0
fuel	4	0	0
town	106	6	6
leasing	2	0	0
condition	2	0	0
air_condition	2	0	0

Feature	Distinct values	Categories with fewer than 3 rows	Rows in rare categories
power_steering	2	0	0
power_mirror	2	0	0
power_window	2	0	0

Many model labels are rare, motivating native categorical handling and regularized one-hot alternatives. Year/price Spearman association is 0.65; engine capacity/price association is 0.07. These are marginal associations, not causal effects or sufficient reasons to remove an interacting feature.

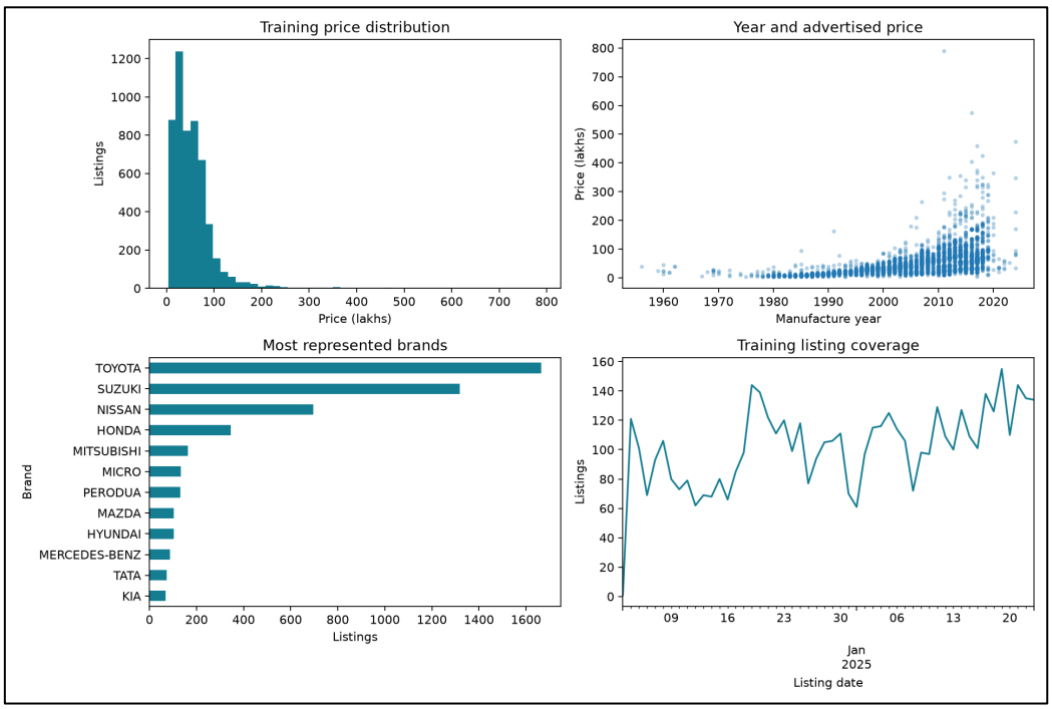


Figure 1: Training-only exploration

4. Data preparation and leakage controls

18 exact duplicate rows were removed after category trimming, and 0 rows failed target/date/year checks. No literal source nulls were found, but the publisher already preprocessed the table; missing original observations and prior imputation cannot be reconstructed. Mileage equals (2025 - year) * 11000 throughout the source and is excluded. The exported index, date, target and group identifiers are not predictors.

The latest record is reserved for a single-example demo, excluding its feature signature from development. Test dates start on 2025-01-24. Matching feature groups are removed from

earlier training data (411 rows). There are 5,285 training, 4,073 test and one demo row. Feature signatures are a conservative proxy; without original listing IDs they cannot establish true vehicle identity. Three expanding temporal folds are used for parameter search, with matching validation groups purged from each training fold.

5. Model selection and preprocessing

- **Random Forest:** bagging reference for nonlinear interactions and variance reduction.
- **XGBoost:** regularized sequential boosting for nonlinear interactions.
- **CatBoost:** native categorical processing is relevant to rare vehicle model labels.
- **LightGBM:** an efficient histogram-based boosting alternative with bounded leaf growth.
- **NGBoost**: shallow trees learn the mean and scale of a price distribution using natural gradients.
- **Ridge:** additive linear comparison to test whether tree complexity adds value.
- **Stacking:** fixed RF and XGBoost pipelines combined through scaled Ridge regression.
- **Training median:** no-feature reference benchmark, not a selectable app model.

All preprocessing is fitted within folds. RF, XGBoost, LightGBM and Ridge use numeric median and categorical mode imputation plus one-hot encoding with rare category pooling. Ridge scales numeric inputs; tree models do not require scaling. CatBoost receives string categories with a missing marker and fold-fitted numeric median imputation. No globally fitted target encoder is used. NGBoost uses the same fold-fitted imputation and rare-category pooling with dense one-hot encoding. All seven estimators use identical outer evaluation partitions.

RF, XGBoost, CatBoost and LightGBM each compare three explicitly constrained configurations. Ridge compares alpha 1, 10 and 100. Actual settings and every search result are in experiments.csv; bounded searches are not exhaustive optimization. CatBoost's adapter supplies categorical names during fit for sklearn compatibility. LightGBM uses ordinary GBDT with row sampling, not a replication of GOSS.

Stacking uses inner purged temporal out-of-fold predictions to train its Ridge meta-model (alpha 10), then refits bases on the training partition. Outer temporal folds evaluate the complete stack. Its bases are fixed, not selected with outer validation labels. Earliest rows without out-of-fold predictions do not train Ridge.

NGBoost formulation and configuration

NGBoost models price conditional on the vehicle inputs as a Normal distribution with mean $\mu(x)$ and scale $\sigma(x)$. Shallow regression trees update both parameters using natural gradients, which adjust ordinary score gradients for the geometry of the distribution. Training minimizes negative log-likelihood (LogScore). Validation MAE selects the configuration, and the application displays the fitted mean as a price estimate. The Normal assumption is a modelling choice rather than a claim that observed prices are Normally distributed (Duan et al., 2020).

NGBoost setting	**Implementation**
Base learners	Depth 3 regression trees; minimum 10 observations per leaf
Fixed settings	Learning rate 0.03; natural gradient enabled; seed 42
Bounded search	100, 200 and 300 rounds across three purged temporal folds
Selected setting	300 rounds, selected by mean validation MAE

The search grid was fixed before viewing NGBoost test results, with no test-driven retuning. NGBoost is evaluated as a separate candidate; the stack continues to combine only Random Forest and XGBoost through its Ridge meta-model.

6. Evaluation and model justification

Mean validation MAE is the selection rule fixed before the expanded run. MAE expresses average monetary error directly; RMSE emphasizes large mistakes and R^2 is unitless, not percentage accuracy. Fold standard deviation is descriptive, not a confidence interval. Timing includes parameter search and refitting on this machine; hardware-specific timings are not published-paper speed comparisons.

Predictive performance

Model	Validation MAE (lakhs)	Fold SD (lakhs)	Test MAE (lakhs)	Test RMSE (lakhs)	Test R^2
Random Forest	7.148	0.146	5.830	14.440	0.893
Stacking	7.299	0.249	6.475	14.411	0.894
XGBoost	7.562	0.312	6.821	14.241	0.896
CatBoost	8.740	0.364	8.414	15.606	0.875
LightGBM	9.410	0.230	7.590	15.283	0.880
NGBoost	9.435	0.461	8.581	16.446	0.861
Ridge	11.124	0.671	9.994	20.694	0.781

Computational performance

Model	Training (s)	Prediction (ms/row)
Random Forest	8.168	0.011
Stacking	10.281	0.015
XGBoost	1.404	0.004
CatBoost	4.529	0.002
LightGBM	1.826	0.007
NGBoost	37.045	0.355
Ridge	0.259	0.003

Model choice and evidence

Random Forest is selected because it has the lowest mean validation MAE (7.15 lakhs) among the evaluated configurations. Test MAE is 5.83 lakhs, compared with 28.67 for a training-median prediction (79.7% lower error on this historical split). This demonstrates improvement over a simple local benchmark, not over external studies.

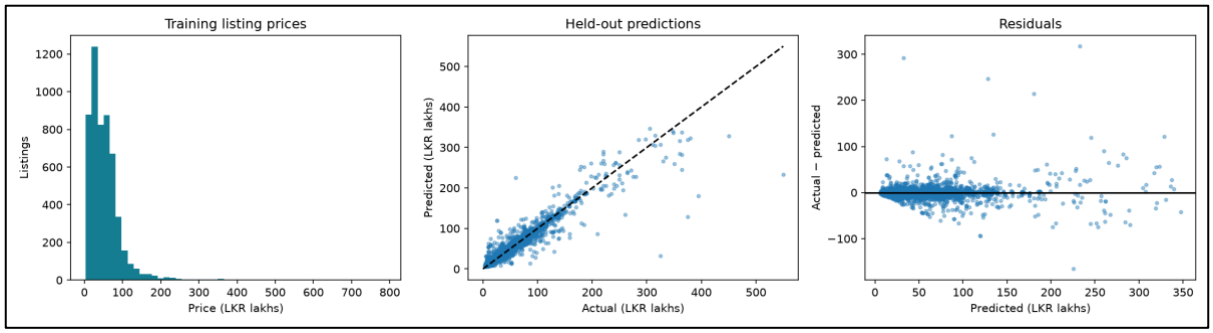


Figure 2: Training listing prices, held-out predictions and residuals

The training data has 1,061 model labels, including 758 occurring fewer than three times. Median training price is 42.5 lakhs and 95th percentile 120.0 lakhs. CatBoost tests native categorical handling; LightGBM tests another efficient nonlinear approach; Ridge provides a simpler additive reference. RF, XGBoost and stacking provide complementary ensemble comparisons. NGBoost tests a distributional approach: it learns both price location and scale, allowing uncertainty evaluation beyond the point-error metrics.

Selection uses MAE because it measures average monetary error directly. RMSE reveals the influence of large mistakes; R^2 is not percentage accuracy. Runtime and fold variation are supporting evidence, not alternate post-hoc selection rules. A more complex model is not automatically better. Error by actual-price band and unseen model label must be considered before trusting an individual estimate.

Limit: this test split was already inspected during earlier development. Expanded results are exploratory. Newly collected future listings are required for independent confirmation. The source covers December 2024-February 2025; estimates are historical asking prices rather than current valuations or verified sale prices.

NGBoost point prediction and uncertainty

NGBoost records validation MAE of 9.43 lakhs (fold SD 0.46), training MAE of 8.20 lakhs, test MAE of 8.58 lakhs, test RMSE of 16.45 lakhs and test R^2 of 0.861. It does not displace Random Forest under the predefined selection criterion. The larger RMSE indicates substantial errors for some listings, while a small training–test MAE gap alone does not establish generalization to a future market.

Distribution measure	NGBoost historical test result
Nominal interval level	90%
Observed interval coverage	89.9% of 4,073 test prices
Mean interval width	33.61 lakhs
Intervals with negative lower bounds	2.4%
Mean negative log-likelihood	3.741 with prices in lakhs

Coverage near 90% does not by itself imply precise or reliable uncertainty. The intervals are wide, and the Normal distribution permits negative prices. These intervals have not been independently calibrated and are not guaranteed ranges for a specific car. The application presents the mean prediction rather than an interval. Recorded search and refit time is 37.04 seconds; mean inference time is 0.355 milliseconds per row on the development machine.

7. Error analysis

The band table below uses actual target prices for retrospective analysis only; those bands are not available when making a new prediction. Counts matter: small subgroups cannot establish reliable group-level performance.

Model	Actual price band (lakhs)	Listings	MAE (lakhs)	RMSE (lakhs)
Random Forest	0–25	938	3.984	7.623
Random Forest	25–50	1307	4.025	7.912
Random Forest	50–100	1467	5.240	9.774
Random Forest	100–200	299	12.406	21.292
Random Forest	>200	62	54.080	83.984
XGBoost	0–25	938	5.191	9.362
XGBoost	25–50	1307	4.984	8.400
XGBoost	50–100	1467	6.295	10.749
XGBoost	100–200	299	14.018	21.792
XGBoost	>200	62	47.974	74.062
CatBoost	0–25	938	5.949	9.824
CatBoost	25–50	1307	5.760	8.970
CatBoost	50–100	1467	7.798	12.323
CatBoost	100–200	299	21.727	27.766
CatBoost	>200	62	52.043	74.383

Model	Actual price band (lakhs)	Listings	MAE (lakhs)	RMSE (lakhs)
LightGBM	0–25	938	5.452	9.402
LightGBM	25–50	1307	5.375	8.885
LightGBM	50–100	1467	7.081	11.588
LightGBM	100–200	299	17.041	24.746
LightGBM	>200	62	53.081	78.813
NGBoost	0–25	938	6.498	10.869
NGBoost	25–50	1307	6.652	10.492
NGBoost	50–100	1467	7.639	11.742
NGBoost	100–200	299	18.299	25.911
NGBoost	>200	62	56.183	84.616
Ridge	0–25	938	9.450	14.718
Ridge	25–50	1307	7.087	13.108
Ridge	50–100	1467	7.426	13.361
Ridge	100–200	299	19.862	26.525
Ridge	>200	62	92.647	116.693
Stacking	0–25	938	4.355	8.304
Stacking	25–50	1307	4.703	8.438
Stacking	50–100	1467	6.173	10.391
Stacking	100–200	299	13.469	22.157
Stacking	>200	62	49.293	78.588

Model	Model-label coverage	Listings	MAE (lakhs)
Random Forest	Seen model label	3484	4.830
Random Forest	Unseen model label	589	11.748
XGBoost	Seen model label	3484	5.993
XGBoost	Unseen model label	589	11.721
CatBoost	Seen model label	3484	7.664
CatBoost	Unseen model label	589	12.851
LightGBM	Seen model label	3484	6.760
LightGBM	Unseen model label	589	12.497

Model	Model-label coverage	Listings	MAE (lakhs)
NGBoost	Seen model label	3484	7.779
NGBoost	Unseen model label	589	13.325
Ridge	Seen model label	3484	8.379
Ridge	Unseen model label	589	19.542
Stacking	Seen model label	3484	5.622
Stacking	Unseen model label	589	11.518

Per-brand counts and MAE for every model are saved in errors_by_brand_all_models.csv. Permutation importance measures how much test MAE changes when an input is shuffled. It explains global reliance, not causation or the reason for a single prediction. Correlated inputs can share importance; shuffling may create implausible vehicles.

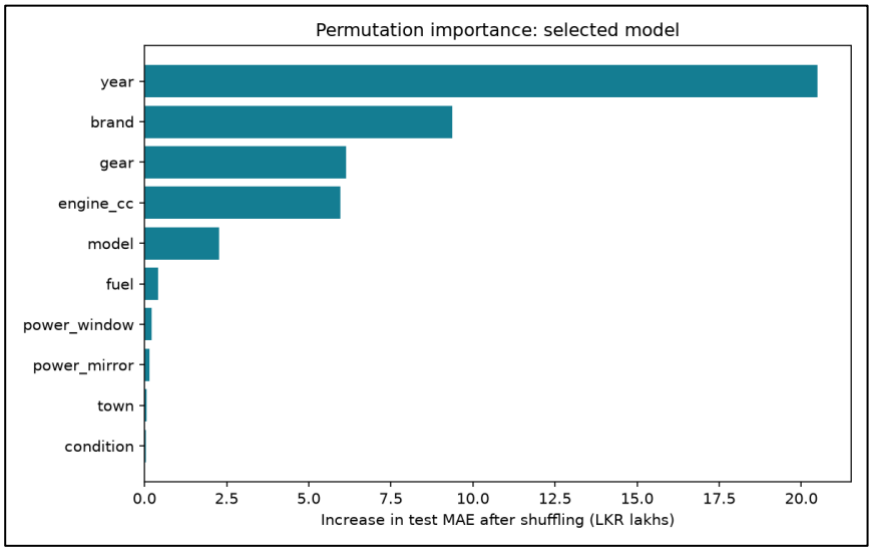


Figure 3: Permutation importance: selected model

NGBoost feature reliance and partial dependence

Feature	Permutation MAE increase (lakhs)
Manufacture year	18.73
Brand	7.13
Engine capacity	5.94
Transmission	5.27

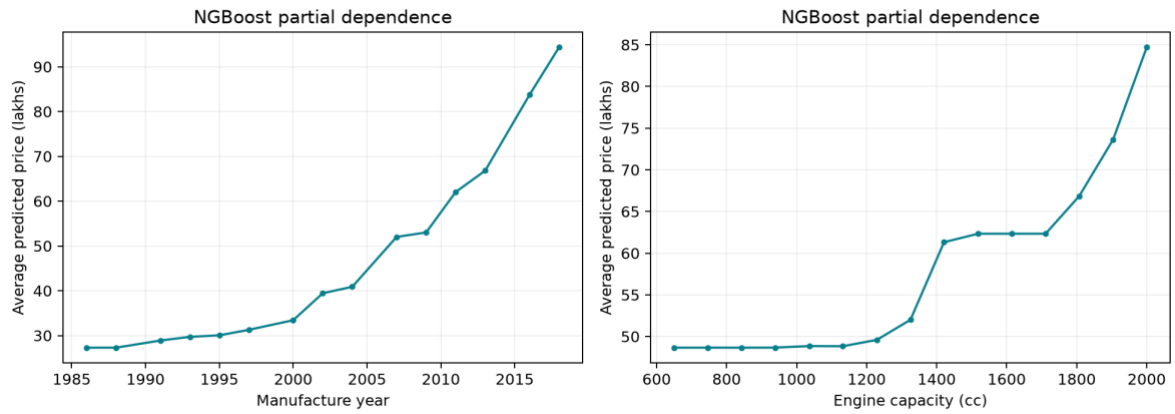


Figure 4: NGBoost partial dependence using training background vehicles

The partial-dependence curves average predictions over 256 sampled training vehicles while varying one field across its training 5th–95th percentile range. From 1986 to 2018, the average prediction changes from 27.28 to 94.33 lakhs; from 650 to 2,000 cc it changes from 48.66 to 84.69 lakhs. Year and brand show the greatest permutation reliance. The broad rising responses are compatible with vehicle age and class differences, but correlated inputs can create implausible combinations. These are learned associations, not causal depreciation curves or explanations of an individual vehicle.

8. Comparison with existing approaches

The methodological comparison below distinguishes published methods from the project's local experiments. Numerical comparisons use identical training, validation and test partitions; published benchmark scores are not treated as directly comparable.

Approach	Source evidence	Comparison with this project	Limit of the comparison
Sri Lankan used-car regression and averaging	Basnayaka (2023) compares linear regression, Random Forest, Extra Trees, gradient boosting, LightGBM and XGBoost, then averages predictions.	We evaluate related linear and tree families; retain RF/XGBoost stacking instead of simple averaging; audit synthetic mileage and use purged temporal evaluation.	The repository abstract reports "accuracy"; its metric and split are not sufficiently specified for numerical equivalence to our MAE/R^2. We do not claim to reproduce or outperform it.

Approach	Source evidence	Comparison with this project	Limit of the comparison
CatBoost	Prokhorenkova et al. (2018) present ordered boosting and categorical-feature processing.	Native categorical input is relevant to our many rare vehicle model labels; encoding is fitted within training folds.	General benchmark results do not guarantee benefits for Sri Lankan vehicle listings. Our configuration uses the library's boosting default; we do not claim to reproduce the paper's experiments.
LightGBM	Ke et al. (2017) describe efficient gradient-boosted trees, including GOSS and feature bundling.	Histogram-based LightGBM is evaluated as an efficiency-oriented nonlinear candidate, with restricted leaves and minimum leaf counts.	Our ordinary GBDT configuration uses row subsampling, not a GOSS replication. Runtime is measured locally; published speedups are not transferred to this dataset.
NGBoost	Duan et al. (2020) introduce natural-gradient boosting for probabilistic prediction.	Normal-distribution regression with shallow trees is evaluated on the same partitions. Coverage and width complement MAE.	The distribution is assumed, and this historical test set was previously inspected. Neither superiority over the paper nor independent calibration is established.
Median and regularized linear references	Implemented here on the same splits.	Median establishes a no-feature error level; Ridge tests whether tree complexity improves on an additive relationship.	These are local benchmarks, not reproductions of the Sri Lankan study's linear model.

9. Single-sample inference and user interface

Reserved source row 9763: actual 30.950 lakhs, selected-model prediction 29.963, absolute error 0.987. All exported estimators are reloaded and checked against their in-memory prediction. The Streamlit interface validates fields, switches models, shows estimates in LKR, and presents model-specific importance, diagnostics, comparison and optional data exploration. For the same reserved ZOTYE Z100 listing, NGBoost predicts approximately LKR 4,929,350 (49.294 lakhs), compared with the actual 30.950 lakhs. This individual

example illustrates disagreement between models and is not evidence of aggregate accuracy. The NGBoost explanation panel also displays partial-dependence curves.

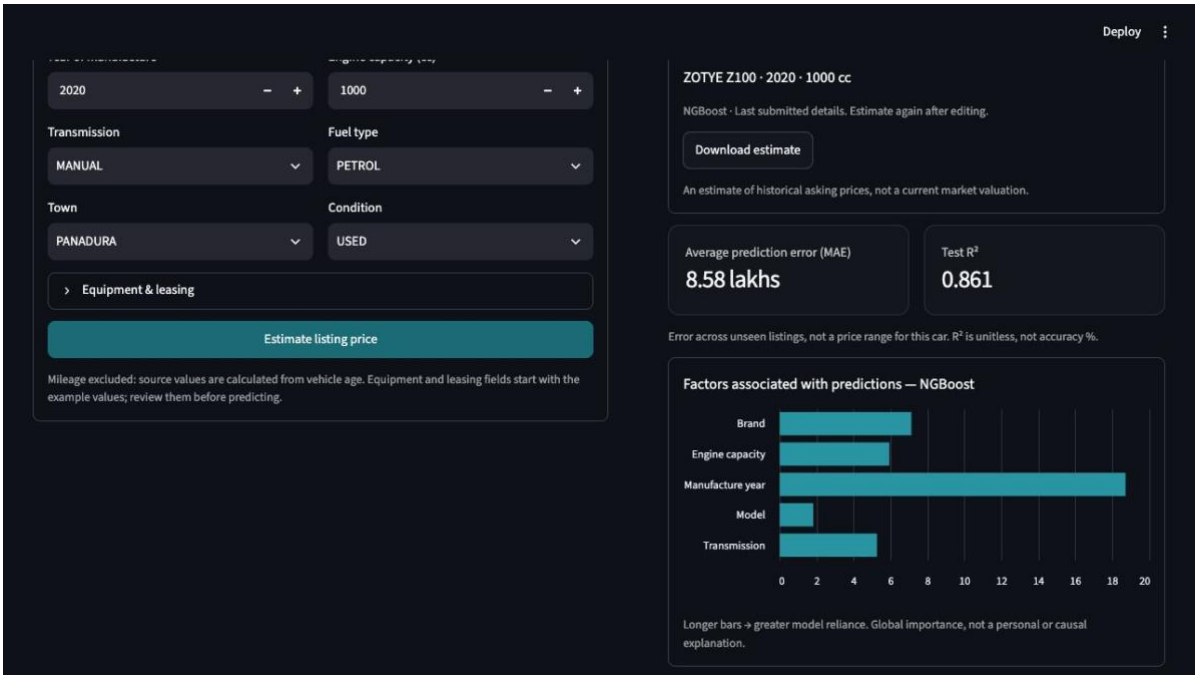


Figure 5: NGBoost performance metrics and feature importance in the application

10. Deployment and repository

The application is deployed on Streamlit Community Cloud and uses the GitHub repository as its deployment source. The entry point is app/app.py. Saved model pipelines and their metadata support inference without retraining when a user changes the selected model.

- Deployed application: Sri Lankan Vehicle Price Prediction on Streamlit
- Source repository: Sri Lankan Vehicle Price Prediction on GitHub

Users enter vehicle characteristics, choose one of the seven trained models and request an estimated listing price in LKR. The interface also provides model comparison metrics, permutation importance, prediction diagnostics and an optional explorer for the raw and processed datasets. Model switching updates the relevant results and explanations. Estimates describe the historical listing data used in this study and should not be interpreted as verified transaction prices. NGBoost additionally supplies training-background partial-dependence plots; distribution intervals are evaluated in the report rather than offered as guaranteed ranges in the prediction form.

The repository contains the application, preprocessing and training code, saved model artifacts, evaluation evidence and the analysis notebook. The README provides installation and reproduction instructions. The interface specification is available in docs/interface.md.

The initial deployment verification on 15 October 2026 confirmed the six model options available at that time and live inference for CatBoost, LightGBM and Ridge. The current seven-model implementation passed 31 local automated tests on 28 October 2026, and the updated notebook executed successfully. Checks cover split isolation, model loading, switching and stacking. Cloud verification of NGBoost remained pending because the browser required sign-in. These checks establish implementation functionality rather than independent predictive validity.

Access follows the application's and repository's sharing settings. Where either resource remains private, the reviewer requires authorized access.

11. Limitations and future evaluation

The historical test set was examined during earlier development, so comparisons involving later candidates remain exploratory. Resampling these rows cannot create an independent test. Future evaluation should freeze model choices before collecting later listings with verified units, observed mileage, vehicle identifiers and, where available, sale prices. Duplicate vehicle identities must be excluded across partitions. The source covers only about two months, is publisher-preprocessed, and may not reflect later import policies, exchange rates or supply changes. NGBoost adds distributional assumptions: Normal intervals can extend below zero, and coverage may change as the market shifts. Independent interval calibration and subgroup coverage checks are required. No causal, fairness, statistical-superiority or current-market accuracy claim is made. No deep learning, computer vision or reinforcement learning is used.

Representation, privacy and responsible use

Online advertisers do not represent the entire Sri Lankan vehicle market. Rare vehicles, expensive listings and regions may have unequal coverage. Brand, town and equipment can reflect geographic or socioeconomic differences; subgroup error tables are not a protected-group fairness audit. Seller names, phone numbers and contact details are not predictors. Public availability alone does not establish consent for every downstream use. Underestimation can disadvantage sellers, while overestimation can mislead buyers. Estimates should support

human judgement and independent inspection, rather than automated lending, insurance or eligibility decisions.

12. Conclusion

The study demonstrates an end-to-end workflow for historical vehicle listing-price regression, from data preparation and temporal evaluation to a deployed inference interface. Random Forest is the preferred model under the predefined validation-MAE criterion. XGBoost achieves a slightly lower test RMSE, while stacking does not improve validation MAE over Random Forest. The expanded comparison therefore supports evidence-based selection rather than assuming that additional model complexity produces better predictions. Across all seven models, NGBoost adds a probabilistic perspective but has higher point error than Random Forest. Its 89.9% coverage of nominal 90% intervals must be read alongside their 33.61-lakh average width and occasional negative bounds. Distributional outputs therefore require validation of usefulness as well as coverage.

Performance varies substantially across price bands and previously unseen vehicle model labels. Reliable evaluation on new listings, with verified vehicle identities and source measurements, is the principal next step before broader practical use.

13. References

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4. CatBoost categorical-feature documentation. Supports using native categorical input rather than external one-hot encoding.
5. LightGBM parameter-tuning documentation. Supports constraining leaves and minimum observations to control complexity.
6. Duan, T., Anand, A., Ding, D. Y., Thai, K. K., Basu, S., Ng, A., and Schuler, A. (2020). NGBoost: Natural Gradient Boosting for Probabilistic Prediction. Proceedings of ICML, PMLR 119:2690–2700. https://proceedings.mlr.press/v119/duan20a.html
7. NGBoost usage documentation. Distributional regression, scoring rules and prediction APIs. https://stanfordmlgroup.github.io/ngboost/useage/

Appendix A. Model-specific strengths, limitations and results

Overview

Random Forest is recommended by validation MAE. All seven models use the same 4,073 held-out listings. MAE measures average error; RMSE emphasizes large errors; R^2 is unitless. See the comparison table for measured results, not claims of statistical superiority.

Additional model candidates and stacking are exploratory additions after the test set was examined. Stacking uses fixed base configurations and a Ridge meta-model trained on inner temporal out-of-fold predictions. Outer temporal folds evaluate the full stacking procedure. The test set is unchanged and was already examined before stacking was added; results are exploratory. All models describe historical asking prices, not completed sales or current valuations.

Random Forest: strengths and limitations

Strengths. Averages many trees to reduce prediction variance. Captures nonlinear relationships and interactions.

Limitations. Rare vehicles can have large errors. Cannot reliably extrapolate beyond training prices.

Measured results. Validation MAE: **7.15 lakhs**; test MAE: **5.83 lakhs**; test RMSE: **14.44 lakhs**; test R^2: **0.893**. Training MAE was **2.68 lakhs**. Lower MAE/RMSE is better; higher R^2 is better. R^2 is unitless, not percentage accuracy.

Feature-importance chart. The strongest fields are **year, brand, gear, engine cc**, in that order. Longer bars mean shuffling that feature increased test error more. This is overall model reliance, not proof of causation, the direction of an effect, or an explanation for a single car. Correlated fields can share importance, and shuffling can produce unrealistic combinations.

Actual-versus-predicted graph. Points on the dashed diagonal are correct predictions. Points below it are underestimates; points above it are overestimates. Each point represents one unseen test listing.

Residual graph. Residual = actual price - predicted price. Positive values mean underestimation; negative values mean overestimation. A tight cloud around zero is desirable. The mean signed residual is **0.08 lakhs**; the largest absolute test error is **317.34 lakhs**. These describe the whole test set, not a guaranteed range for the current vehicle.

What the graphs show here. Most listings form a dense cluster at the lower end of the price range, broadly following the diagonal. The diagnostics show the direction and size of errors for this model. Points far from the diagonal or zero line identify large errors; no model is guaranteed reliable across all vehicles. The plots alone do not establish why those listings were difficult.

Large-error interpretation. RMSE (14.44) is much larger than MAE (5.83), consistent with some large errors having a strong effect on the squared-error metric. Inspect isolated points far from the diagonal/zero line; an average score can hide these failures.

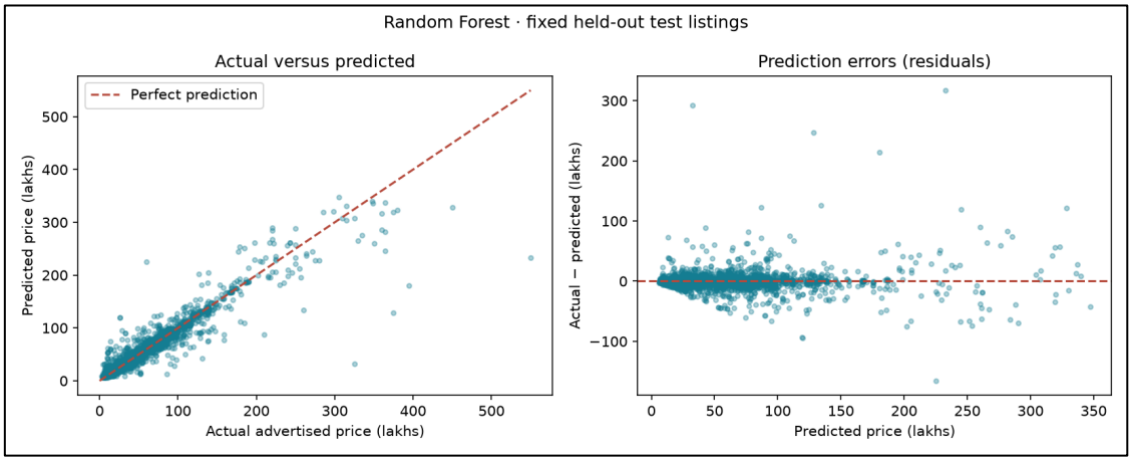


Figure 6: Random Forest diagnostic graphs

Stacking: strengths and limitations

Strengths. Combines Random Forest and XGBoost predictions using Ridge. Meta-model learns from later out-of-fold predictions.

Limitations. Similar base-model errors can limit improvement. More training work; early rows are excluded from meta-training.

Measured results. Validation MAE: **7.30 lakhs**; test MAE: **6.47 lakhs**; test RMSE: **14.41 lakhs**; test R^2: **0.894**. Training MAE was **4.78 lakhs**. Lower MAE/RMSE is better; higher R^2 is better. R^2 is unitless, not percentage accuracy.

Feature-importance chart. The strongest fields are **year, brand, engine cc, gear**, in that order. Longer bars mean shuffling that feature increased test error more. This is overall model reliance, not proof of causation, the direction of an effect, or an explanation for a single car. Correlated fields can share importance, and shuffling can produce unrealistic combinations.

Actual-versus-predicted graph. Points on the dashed diagonal are correct predictions. Points below it are underestimates; points above it are overestimates. Each point represents one unseen test listing.

Residual graph. Residual = actual price - predicted price. Positive values mean underestimation; negative values mean overestimation. A tight cloud around zero is desirable. The mean signed residual is **-0.90 lakhs**; the largest absolute test error is **296.06 lakhs**. These describe the whole test set, not a guaranteed range for the current vehicle.

What the graphs show here. Most listings form a dense cluster at the lower end of the price range, broadly following the diagonal. The diagnostics show the direction and size of errors for this model. Points far from the diagonal or zero line identify large errors; no model is guaranteed reliable across all vehicles. The plots alone do not establish why those listings were difficult.

Large-error interpretation. RMSE (14.41) is much larger than MAE (6.47), consistent with some large errors having a strong effect on the squared-error metric. Inspect isolated points far from the diagonal/zero line; an average score can hide these failures.

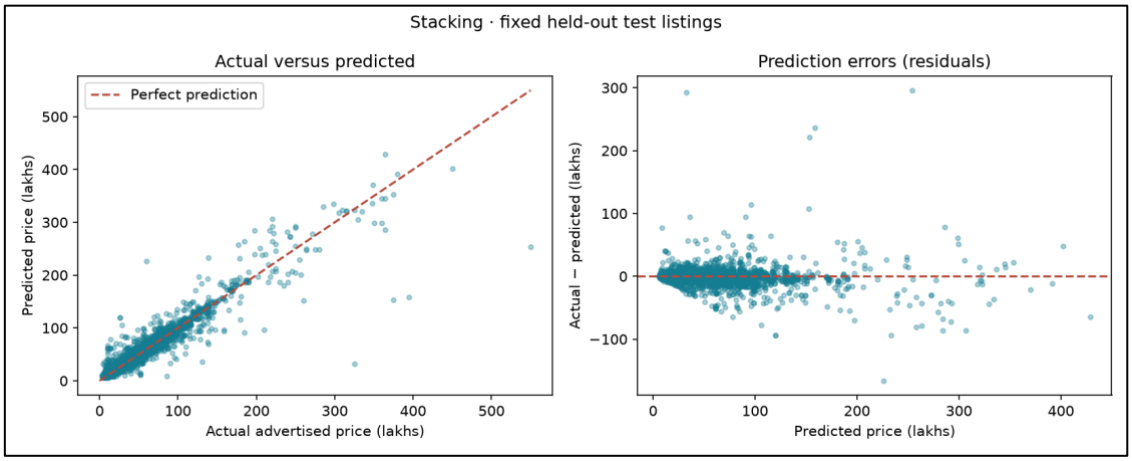


Figure 7: Stacking diagnostic graphs

XGBoost: strengths and limitations

Strengths. Successive trees correct earlier errors. Regularization controls complex nonlinear patterns.

Limitations. Sensitive to parameter choices. Rare categories and market changes can still cause large errors.

Measured results. Validation MAE: **7.56 lakhs**; test MAE: **6.82 lakhs**; test RMSE: **14.24 lakhs**; test R^2: **0.896**. Training MAE was **5.57 lakhs**. Lower MAE/RMSE is better; higher R^2 is better. R^2 is unitless, not percentage accuracy.

Feature-importance chart. The strongest fields are **year, brand, engine cc, gear**, in that order. Longer bars mean shuffling that feature increased test error more. This is overall model reliance, not proof of causation, the direction of an effect, or an explanation for a single car. Correlated fields can share importance, and shuffling can produce unrealistic combinations.

Actual-versus-predicted graph. Points on the dashed diagonal are correct predictions. Points below it are underestimates; points above it are overestimates. Each point represents one unseen test listing.

Residual graph. Residual = actual price - predicted price. Positive values mean underestimation; negative values mean overestimation. A tight cloud around zero is desirable. The mean signed residual is **-0.24 lakhs**; the largest absolute test error is **294.56 lakhs**. These describe the whole test set, not a guaranteed range for the current vehicle.

What the graphs show here. Most listings form a dense cluster at the lower end of the price range, broadly following the diagonal. The diagnostics show the direction and size of errors for this model. Points far from the diagonal or zero line identify large errors; no model is guaranteed reliable across all vehicles. The plots alone do not establish why those listings were difficult.

Large-error interpretation. RMSE (14.24) is much larger than MAE (6.82), consistent with some large errors having a strong effect on the squared-error metric. Inspect isolated points far from the diagonal/zero line; an average score can hide these failures.

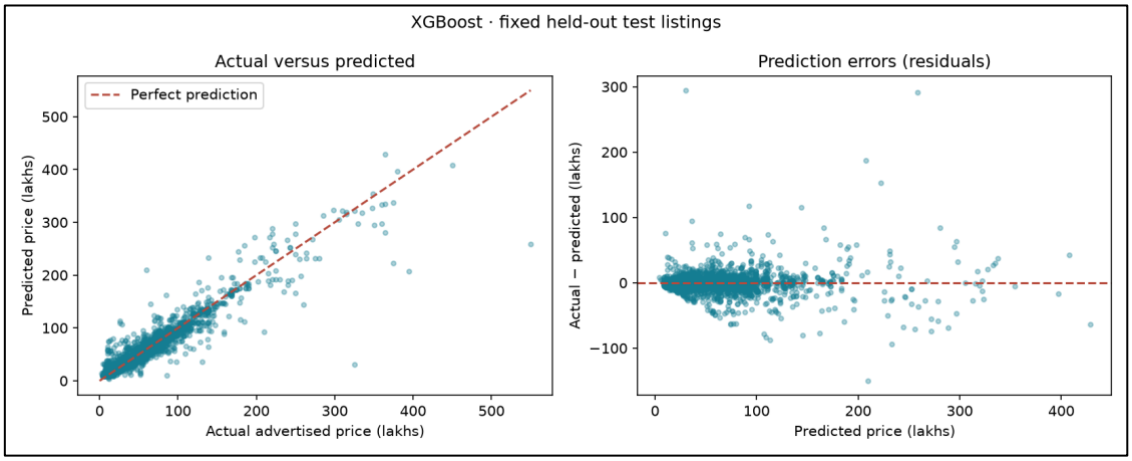


Figure 8: XGBoost diagnostic graphs

CatBoost: strengths and limitations

Strengths. Uses categorical fields directly, without external one-hot encoding. Can capture patterns across many vehicle model categories.

Limitations. Rare categories still have limited supporting evidence. Categorical statistics must be learned inside training folds.

Measured results. Validation MAE: **8.74 lakhs**; test MAE: **8.41 lakhs**; test RMSE: **15.61 lakhs**; test R^2: **0.875**. Training MAE was **7.76 lakhs**. Lower MAE/RMSE is better; higher R^2 is better. R^2 is unitless, not percentage accuracy.

Feature-importance chart. The strongest fields are **year, brand, engine cc, gear**, in that order. Longer bars mean shuffling that feature increased test error more. This is overall model reliance, not proof of causation, the direction of an effect, or an explanation for a single car. Correlated fields can share importance, and shuffling can produce unrealistic combinations.

Actual-versus-predicted graph. Points on the dashed diagonal are correct predictions. Points below it are underestimates; points above it are overestimates. Each point represents one unseen test listing.

Residual graph. Residual = actual price - predicted price. Positive values mean underestimation; negative values mean overestimation. A tight cloud around zero is desirable. The mean signed residual is **0.11 lakhs**; the largest absolute test error is **302.58 lakhs**. These describe the whole test set, not a guaranteed range for the current vehicle.

What the graphs show here. Most listings form a dense cluster at the lower end of the price range, broadly following the diagonal. The diagnostics show the direction and size of errors for this model. Points far from the diagonal or zero line identify large errors; no model is guaranteed reliable across all vehicles. The plots alone do not establish why those listings were difficult.

Large-error interpretation. RMSE (15.61) is much larger than MAE (8.41), consistent with some large errors having a strong effect on the squared-error metric. Inspect isolated points far from the diagonal/zero line; an average score can hide these failures.

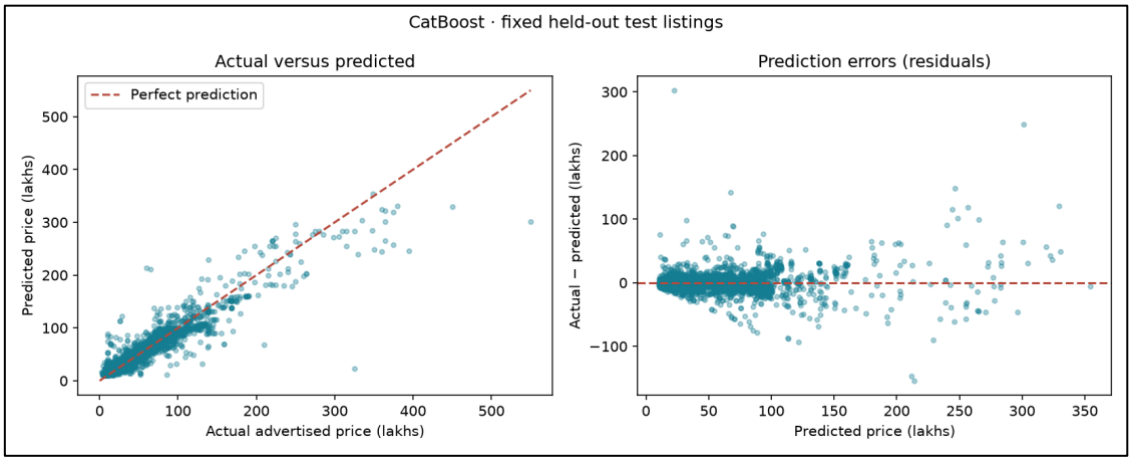


Figure 9: CatBoost diagnostic graphs

LightGBM: strengths and limitations

Strengths. Histogram-based boosting is an efficient tabular candidate. Leaf-wise trees capture nonlinear relationships.

Limitations. Leaf growth can overfit small groups without constraints. This implementation uses one-hot categories; unseen labels provide limited information.

Measured results. Validation MAE: **9.41 lakhs**; test MAE: **7.59 lakhs**; test RMSE: **15.28 lakhs**; test R^2: **0.880**. Training MAE was **7.14 lakhs**. Lower MAE/RMSE is better; higher R^2 is better. R^2 is unitless, not percentage accuracy.

Feature-importance chart. The strongest fields are **year, brand, engine cc, gear**, in that order. Longer bars mean shuffling that feature increased test error more. This is overall model reliance, not proof of causation, the direction of an effect, or an explanation for a single car. Correlated fields can share importance, and shuffling can produce unrealistic combinations.

Actual-versus-predicted graph. Points on the dashed diagonal are correct predictions. Points below it are underestimates; points above it are overestimates. Each point represents one unseen test listing.

Residual graph. Residual = actual price - predicted price. Positive values mean underestimation; negative values mean overestimation. A tight cloud around zero is desirable. The mean signed residual is **0.02 lakhs**; the largest absolute test error is **293.00 lakhs**. These describe the whole test set, not a guaranteed range for the current vehicle.

What the graphs show here. Most listings form a dense cluster at the lower end of the price range, broadly following the diagonal. The diagnostics show the direction and size of errors for this model. Points far from the diagonal or zero line identify large errors; no model is guaranteed reliable across all vehicles. The plots alone do not establish why those listings were difficult.

Large-error interpretation. RMSE (15.28) is much larger than MAE (7.59), consistent with some large errors having a strong effect on the squared-error metric. Inspect isolated points far from the diagonal/zero line; an average score can hide these failures.

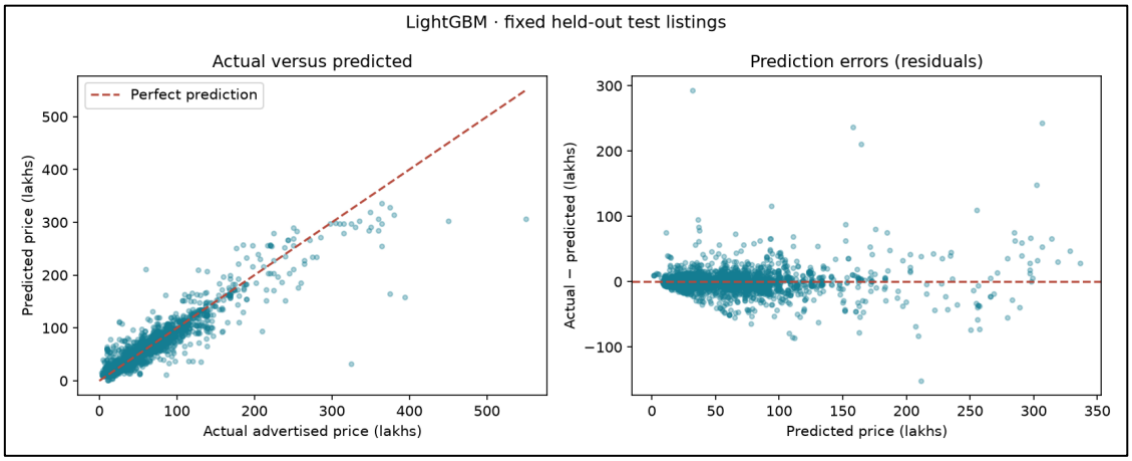


Figure 10: LightGBM diagnostic graphs

NGBoost: strengths and limitations

Strengths. Learns both the mean and scale of a conditional price distribution. Shallow trees capture nonlinear relationships, while natural gradients support joint distribution-parameter updates.

Limitations. The Normal assumption permits negative prices and may poorly describe skewness or outliers. Wide intervals and market shifts limit uncertainty usefulness. NGBoost does not improve validation MAE over Random Forest.

Measured results. Validation MAE: 9.43 lakhs; test MAE: 8.58 lakhs; test RMSE: 16.45 lakhs; test R^2: 0.861. Training MAE: 8.20 lakhs. R^2 is unitless, not percentage accuracy.

Feature importance. Year, brand, engine capacity and transmission have the greatest permutation reliance. The partial-dependence curves in Section 7 describe average fitted associations, not causal or individual explanations.

Graph interpretation. Points close to the diagonal have smaller errors. Residual equals actual minus predicted price: positive values indicate underestimation. The mean residual is −0.19 lakhs and the largest absolute error is 313.17 lakhs. Isolated points far from the reference lines show large failures that average metrics can hide.

Uncertainty interpretation. Nominal 90% intervals cover 89.9% of historical prices, with mean width 33.61 lakhs. The 2.4% negative lower bounds expose the support limitation of the Normal model. These results do not guarantee calibrated ranges for future vehicles.

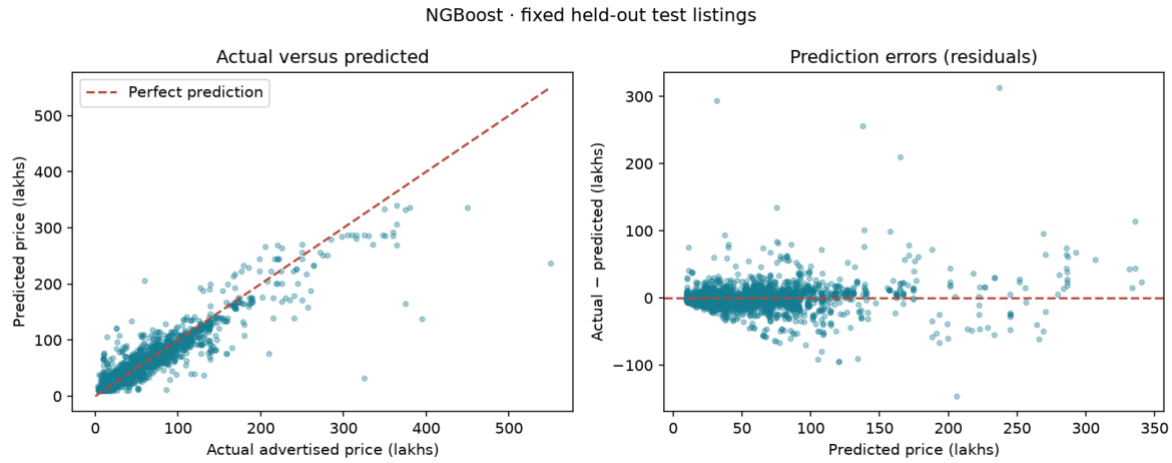


Figure 11: NGBoost diagnostic graphs

Ridge: strengths and limitations

Strengths. Regularized linear reference for measuring the benefit of tree models. Simple additive structure and relatively inexpensive inference.

Limitations. Does not automatically learn nonlinear interactions. Extrapolation can produce unrealistic or negative prices.

Measured results. Validation MAE: **11.12 lakhs**; test MAE: **9.99 lakhs**; test RMSE: **20.69 lakhs**; test R^2: **0.781**. Training MAE was **9.23 lakhs**. Lower MAE/RMSE is better; higher R^2 is better. R^2 is unitless, not percentage accuracy.

Feature-importance chart. The strongest fields are **year, brand, model, engine cc**, in that order. Longer bars mean shuffling that feature increased test error more. This is overall model reliance, not proof of causation, the direction of an effect, or an explanation for a single car. Correlated fields can share importance, and shuffling can produce unrealistic combinations.

Actual-versus-predicted graph. Points on the dashed diagonal are correct predictions. Points below it are underestimates; points above it are overestimates. Each point represents one unseen test listing.

Residual graph. Residual = actual price - predicted price. Positive values mean underestimation; negative values mean overestimation. A tight cloud around zero is desirable. The mean signed residual is **-0.39 lakhs**; the largest absolute test error is **360.61 lakhs**. These describe the whole test set, not a guaranteed range for the current vehicle.

What the graphs show here. Most listings form a dense cluster at the lower end of the price range, broadly following the diagonal. The diagnostics show the direction and size of errors for this model. Points far from the diagonal or zero line identify large errors; no model is guaranteed reliable across all vehicles. The plots alone do not establish why those listings were difficult.

Large-error interpretation. RMSE (20.69) is much larger than MAE (9.99), consistent with some large errors having a strong effect on the squared-error metric. Inspect isolated points far from the diagonal/zero line; an average score can hide these failures.

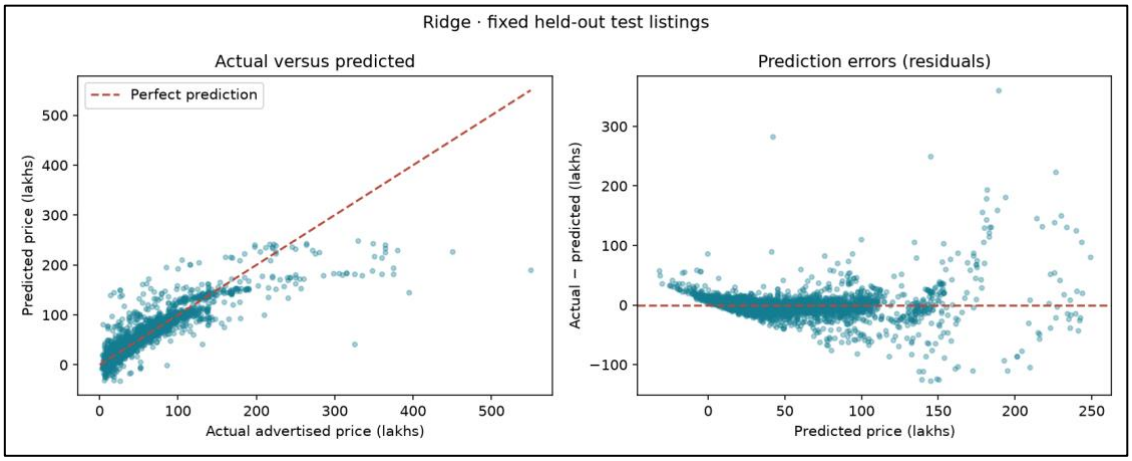


Figure 12: Ridge diagnostic graphs